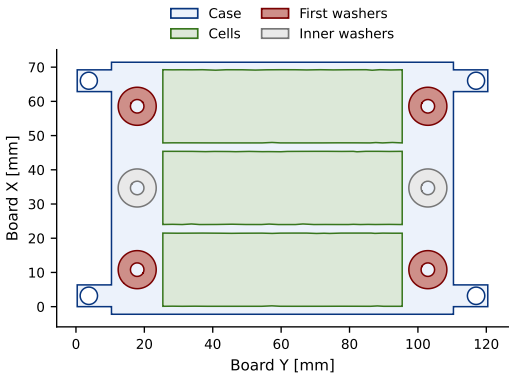
The outer washers meet the case before the cells



Plan view through the cover. Red marks the first four loaded washers.

Lowering the case from 3 mm

The four outer washers touch after 0.603 mm of travel. The rigid tabs are still 2.397 mm from the PCB. The first cell touches after 1.760 mm.

Flush installation

With nylon screws, the flexible-board model crossed the PLA through-layer reference at 0.95–1.10 mm of travel. At 2.75 mm all four tabs touched, yet only 4.4–5.1% of each tab footprint was within 0.05 mm of the PCB. The cells still had clearance.

The hole pattern leaves little room for an M4 screw

	PCB	Case
Bore diameter [mm]	4.200	5.000–5.090
Hole pitch X [mm]	62.865	62.855
Hole pitch Y [mm]	114.427	113.241

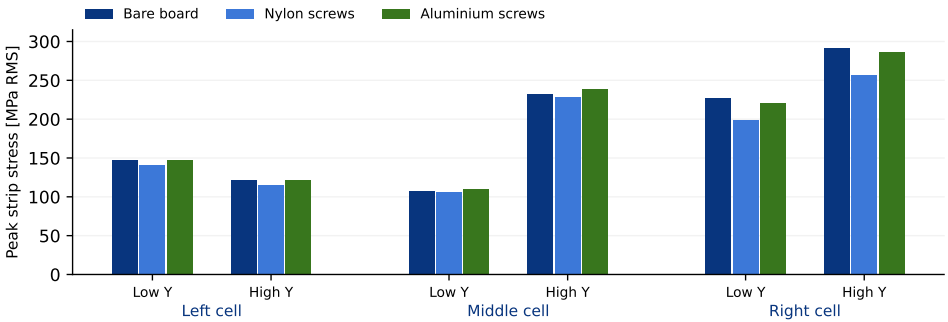
Each end of the centred pattern is about 0.593 mm off-axis. The smaller bore pair allows 0.600 mm with a nominal M4 shank.

Hardware under study

Four M4 case screws in aluminium or nylon. The nylon reference is McMaster-Carr 93640A145.

The dense PLA case adds about 74 g to a 252 g bare assembly. Print orientation and infill affect its installed stiffness and mass.

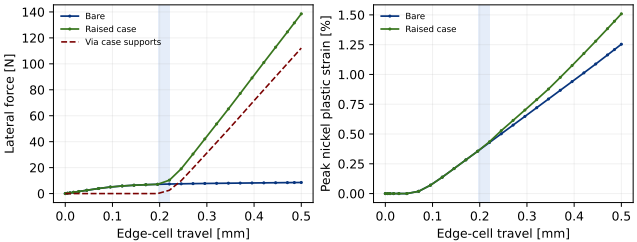
The raised nylon case lowers vibration stress in all six nickel strips



With base X excitation, the highest strip stress was **291.9 MPa RMS** bare, **256.2** with nylon screws and **286.5** with aluminium. All three maxima occurred at the same corner of the right-hand cell attachment.

In the seeded history, all three elastic models crossed the local pull study's assumed 185 MPa yield stress within 39 ms. Both raised cases still had clearance at the cells. Actual nickel fatigue life remains open.

The raised nylon-screw case restrains the edge cell



The blue band brackets first contact at 0.196–0.221 mm of travel.

At the same 8.54 N pull

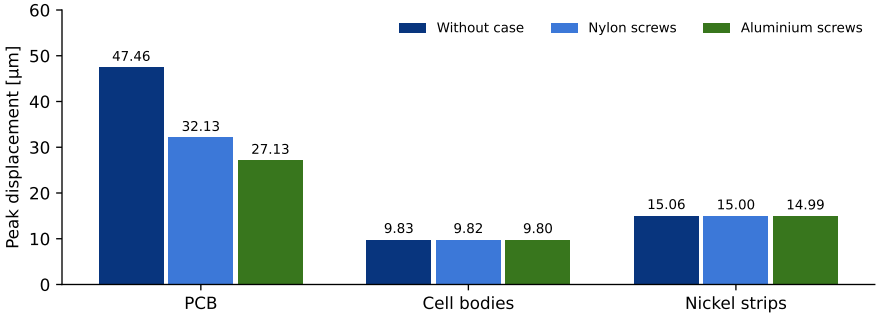
Cell travel fell from 0.50 mm to about 0.21 mm. Peak nickel plastic strain was 1.25% bare; the two cased states bracketing 8.54 N gave 0.36% and 0.43%.

At the same 0.50 mm travel

The pull rose from 8.54 to 138.53 N. The four case screw supports carried 112.15 N. Peak nickel plastic strain rose from 1.25% to 1.51%.

Local fixture, 3 mm stand-off, positive X pull with fixed screw tops.

Aluminium screws further reduced PCB deflection during tightening

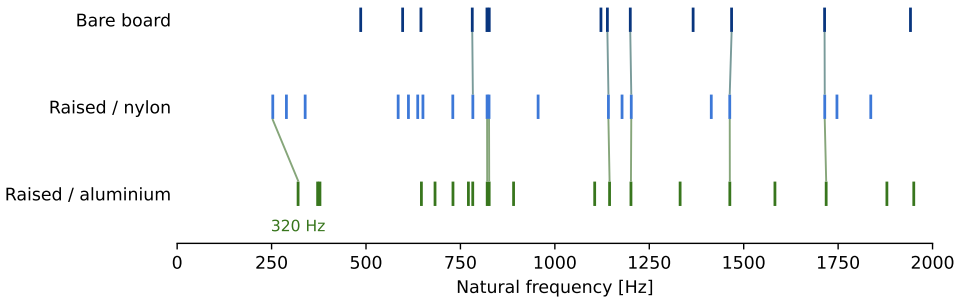


The same PCB node moved 47.46 μm bare, 32.13 μm with nylon screws and **27.13 μm** with aluminium screws. The reductions from bare were 32% and 43%, respectively.

No cell, strip or washer contacted the case during tightening. Load transfer was through the four corner attachments.

3 mm stand-off. Six original bolts loaded to 750 N each and locked. No imposed vibration.

Both raised cases introduce modes below 486 Hz

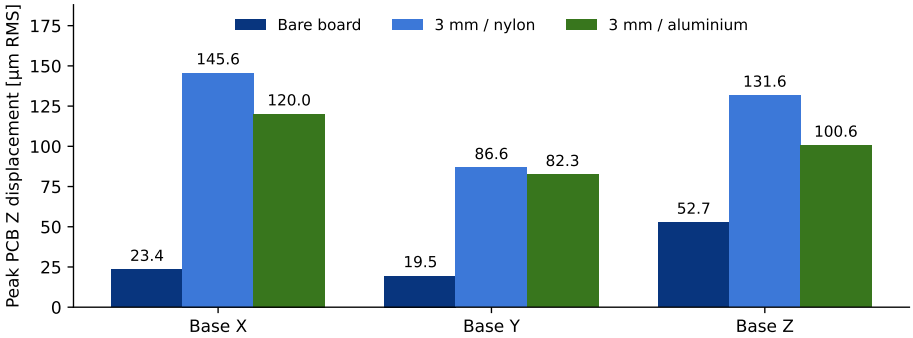


The first cased mode rises from **253 Hz with nylon** to **320 Hz with aluminium**. Its shape remains closely matched across 399,194 solid nodes (MAC 0.989).

The 781 Hz bare battery-motion mode reappears at 783 Hz with the case (shape correlation 0.943). Comparing modes by rank alone would miss this match.

Bolt-locked flexible assemblies at 3 mm stand-off. Undamped mode shapes about the tightened state.

The open-gap vibration model predicts more PCB motion



With base X excitation, peak PCB Z motion was **23.37** μm RMS bare, **145.55** with nylon and **120.00** with aluminium. The figure uses independent modal integration.

Native MAPDL confirms both base-X results. Aluminium reduced board motion relative to nylon, but raised peak stress across the PLA layers near a screw bearing from 17.16 to 24.78 MPa RMS. Both models reached cell contact during the full X history. In the 480-frame clip, only the nylon model crossed the gap (frames 253–270, counting from 1).